

Network traffic classification using convolutional neural network and ant-lion optimization[☆]

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ABSTRACT

Traffic identification has become a challenging task in recent years. Recently, deep learning methods have been extensively studied for network traffic classification recently. Unfortunately, these models require a large amount of training data. Another challenge with most traffic classification methods is that the features must be extracted by an expert. In these methods, finding the desired features that lead to a better classification is very tedious and time-consuming. In this regard, an excellent solution to address this challenge is to use of deep learning methods that automatically extract features. This paper is an attempt to address these issues by using a combination of the convolutional neural network (CNN), the ant-lion meta-heuristic algorithm (ALO), and the self-organizing map (SOM) to create a traffic classification model that can accurately identify traffic types. The proposed method was able to identify encrypted traffic and distinguish between VPN and non-VPN traffics. The model was evaluated using the “ISCX VPN-non-VPN” dataset, and achieved 98% accuracy.

1. Introduction

Network traffic classification is a process for identifying different types of applications [1]. Traffic classification is used in various applications, such as network management, network design, and network security. Network traffic classification is an essential and challenging task for different reasons such as (a) the identification of some applications is complicated because of complex, and dynamic protocols (e.g., P2P, online game, video streaming, etc.) [2,3]; (b) the use of dynamic ports and tunnels, which increases the complexity of traffic identification; and (c) the use of encryption methods that can make it difficult to identify the traffic. Therefore, the classification of encrypted network traffic has become a big challenge in modern networks. It should be noted that many network traffic classification techniques like port number, payload based, statistical methods, and machine learning approaches require patterns or extracted features. This process is time-consuming, error-prone, and expensive.

Network traffic classification using traditional machine learning techniques requires the manual involvement of human experts in the feature extraction process. The manual process is prone to human error, which in recent years has emphasized the need for deep learning-based methods for automatic feature extraction. However, these methods require a large amount of data in the training phase. Since these models contain several hidden layers, the optimized adjustment of weights cannot be performed entirely and correctly with a small amount of data. Researchers have shown that the optimal and desirable result cannot be achieved without a sufficient data volume, and inadequate training data results in a model with poor performance.

However, the performance of traffic classification has been considerably improved by features that can be extracted and learned through deep network models. The problems of robust feature selection and sufficient data availability for some traffics are still

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the two significant challenges. These challenges generally degrade the performance of network traffic classification. For classifying network traffic, deep learning algorithms mainly use an end-to-end deep learning-based classification method, which does not require manual features extraction and also does not provide the opportunity for further optimization of the algorithm. To improve the performance of the network traffic classifier, in addition to the classification performance of the algorithm, the algorithm must be further optimized throughout the designing phase. Therefore, to optimize the algorithm and overcome some of the problems and challenges of deep learning, meta-heuristic algorithms were used. The reason for using them was the fact that they use basic mechanisms and operations to finally achieve a proper and optimal solution through a set of iterations. Ant-lion meta-heuristic algorithm (ALO) is a recently proposed bio-inspired optimization method and one of the meta-heuristic algorithms that shows good performance in terms of searching the feature space for an optimal subset of features [4–8]. The ALO yields very ambitious results in terms of exploitation, improved exploration, local optimum avoidance and convergence in different forms of optimization functions; so that it is widely used for solving problems in different areas [7,8].

The CNN is a type of multi-layer neural networks. Several applications use feed-forward neural networks that involve the back-propagation (BP) learning method. These types of networks use the back-propagation (BP) gradient descent method for training. This causes some problems for the algorithm because the algorithm is based on the gradient. One of the most critical problems is to fall in the local minimum because of the slow convergence speed. Our proposed approach can provide an appropriate and effective solution for these challenges [9].

This paper presents an approach that can identify network traffic, extract features automatically, and solve the problem of insufficient data availability. Our proposed method can classify different types of traffic with high accuracy and, acceptable performance. It can also fix problems, including encrypted packets, dynamic protocols such as P2P (Peer to Peer), and those protocols that use virtual private networks. Therefore, the method presented in this paper can determine the types of applications (protocols) with high accuracy, using a combination of CNN, ALO, and fuzzy-SOM neural network-based clustering.

The essential and beneficial points of the proposed method are:

- Preprocessing traffic data using entropy and entropy variance.
- Automated feature extraction from the hidden layers of the one-dimensional convolutional neural network (1D-CNN).
- Efficient feature selection with high classification accuracy using ALO.
- Classification of new instances using a fuzzy-SOM based-clustering.

This paper is organized in following sections. In Section 2, a literature review is performed and the basic concepts of CNN and meta-heuristic algorithms are introduced in Section 3. Section 4 explains the proposed method and in Section 5, the dataset, model evaluation methods, results, and evaluation are presented. Section 6 concludes the paper.

2. Related works

This section provides an overview of the main methods used to classify network traffic. In general, these methods can be categorized into two main groups: (1) machine learning methods and (2) deep learning methods.

Machine learning methods: There are many studies on traditional machine learning techniques, some of which are briefly described here. Shafiq et al. [10] used various machine learning classifiers such as C4.5 decision trees, Naïve Bayes, Support Vector Machine (SVM), and Bayes Net classifiers for network traffic classification using a real-time Internet dataset. They showed that C4.5 gave a better result (78.9%) in comparison to other machine learning approaches. In [11], Soysal et al. employed three machine learning classifiers, decision trees, Bayesian networks, and multi-layer perceptrons (MLP), using flow traces for traffic classification. The dependence of the performance of network traffic classification systems on the amount of the training data show that machine learning algorithms such as decision trees, and Bayesian networks are suitable for the classification of high-speed Internet traffic flows. They also encompass applications with dynamic port number changing. These algorithms analyze the impact of mislabeled data (training and testing) on the accuracy of machine learning algorithms.

Deep learning methods: Recently, researchers have tried to use deep learning for network traffic. Lixin, et al. [12] proposed a new approach to train a CNN model by optimizing a new function, which emphasized on the learned features for network traffic classification. They evaluated the proposed method on three traffic datasets. The experimental results showed that the method improved the classification accuracy, and it can be well adapted to the challenges of class diversity and class similarity in network traffic classification. Hong, et al. [13] proposed a method to improve network traffic classification performance using deep learning methods. They presented a network applications classification model based on Deep Belief Network (DBN) and used the BP-based model for comparison in the same conditions. The result showed that classifying network traffic based on DBN had a higher accuracy.

Gil et al. [14] used statistical features such as flow arrival time, the number of flows, Bytes per second, etc., to classify network traffic. They also applied a decision tree and KNN algorithms and achieved 92% accuracy for recognizing six classes (chat, email, FTP, etc.). Yamansavascilar et al. [15] considered the “ISCX VPN-non VPN” dataset and 111 flow features with 12 application classes and used the KNN algorithm to achieve 94% accuracy. The main disadvantage of these methods is that an expert does the feature extraction and selection processes. Therefore, such methods are prone to human error, time-consuming, and expensive. In [16], Rezaei et al. presented a general framework to classify network traffic based on deep learning. Lotfollahi et al. [17] introduced a model based on CNN and Stacked Auto-Encoder (SAE) for classifying encrypted traffic. The numbers of classes and data instances in their work were close to those used in the present study. They obtained an accuracy of 94% in traffic classification by the CNN

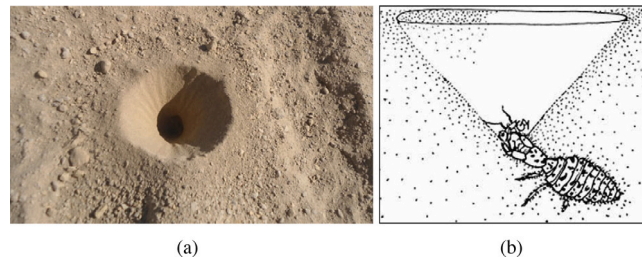


Fig. 1. Tapered hole and an ant-lion hunting behavior. (a) A cone-shaped pit dug by an ant-lion. (b) Waiting in the pit.

model. Wang et al. [18] proposed a method for classifying encrypted traffic using 1D-CNNs. This method combines feature extraction, feature selection, and the classifier into a final unified framework for automated learning of the non-linear relation between the row input and the expected output. Their method applied the “ISCX VPN-non VPN” dataset and obtained an accuracy of 92% for 12 application classes. However, they only used a few instances of the dataset, which is insufficient to draw complete conclusion.

Our proposed method achieved a significant accuracy on the same dataset, using the combination of the 1D-CNN, ALO, and SMO algorithm compared with others work on “ISCX VPN-non VPN” dataset.

3. Convolutional neural network and ant-lion optimization algorithm concepts

In this section, the concepts of the convolutional neural network (CNN) and the ant-lion optimization algorithm (ALO) are described.

3.1. The concept of convolutional neural network

One of the most popular deep learning algorithms is the convolutional neural networks (CNNs). A CNN consists of an input layer, an output layer, and several hidden layers. Mainly, a CNN is composed of three types of layers, namely, convolutional layers, pooling layers, and fully connected layers. In general, CNNs are hierarchical neural networks, in which the convolutional layers and pooling layers are placed alternately, and then multiple fully connected layers are arranged. Performing a convolution operation has three merits [9] (1) using a weight sharing mechanism in every feature map considerably decreases the number of parameters, (2) the local connection learns the relations between neighboring pixels, and (3) it leads to immutability and stability against object replacement. A pooling layer is usually placed after the convolutional layers and can be used to reduce the size of the feature map and the number of network parameters. Then, fully connected layers allow to represent the network’s result as a vector with a defined size. We believe that this method is an optimal option for network traffic classification. The results approved this claim and showed that 1D-CNN performs well in terms of feature extraction from network traffic classification.

3.2. Ant-lion optimizer algorithm

In recent years, meta-heuristic algorithms have been used to obtain optimized solutions for optimization problems. Due to the random nature of these algorithms, the probability of avoiding local optimums is high. If an evolutionary algorithm (EA) traps in a local optimum, then the random operators make random changes in the solution and consequently depart from that local optimum. The EAs use a general and common framework, in which a set of random solutions are created gradually or maybe iteratively. Some of the most famous algorithms in the EA family include the Ant Colony Optimization (ACO), Emperor Penguins Colony algorithm, Artificial Bee Colony optimization, Particle Swarm Optimization (PSO), and Genetic Algorithm (GA).

The ant-lion optimizer (ALO) algorithm was used in this study. The life cycle of an ant-lion consists of two main steps; larva and adulthood. An ant-lion larva digs a cone-shaped hole with its hand by moving in a circular orbit and throws out the soil with its jaw. Fig. 1(a) shows a tapered hole. After digging the trap, the larva hides beneath the cone and waits for insects (ideally ants) to get trapped in the hole, like what is shown in Fig. 1(b) [19]. The edge of the cone is sharp enough so that ants easily fall into the bottom of the trap. As soon as the ant-lion finds out about a miserable prey trapped in the hole, he tries to catch it. However, insects are usually not easy to catch, while trying to fly out of the trap. In this condition, the ant-lion throws soil around the edge, so that the prey slips downward. Once the larva captures the victim with his jaw, it pulls the hung under the soil. After that, the ant-lion throws the rest of the ant’s body out of the hole and prepares the trap for another hunt.

3.2.1. Ant-lion optimization algorithm operators

In the ALO algorithm, the behavior of ant-lions as hunters and ants as prey is modeled [20]. To model this encounter, the ants have to move around the search space, and the ant-lions hunt them. The ant-lion's fitness increases as they set up more traps. Since the ants move randomly in the nature to find foods, a random move for modeling ant movements can be described based on Eq. (1), and Eq. (2) :

$$x(t) = [0, totalsum(2r(t_1) - 1), \dots, totalsum(2r(t_n) - 1)] \quad (1)$$

$$r(t) = \begin{cases} 1 & \text{if } rand > 0.5 \\ 0 & \text{if } rand \leq 0.5 \end{cases} \quad (2)$$

Where *totalsum* shows the total density, *n* indicates the maximum number of iterations; *t* is the random movement step. In Eq. (2), *r(t)* indicates the random function and *rand* shows a random value generated with uniform distribution in the range [0, 1].

The location of the ants is stored in the matrix (3), which is used later in the optimization:

$$M_{Ant} = \begin{bmatrix} A_{1,1} & A_{1,2} & \dots & A_{1,d} \\ A_{2,1} & A_{2,2} & \dots & A_{2,d} \\ \vdots & \vdots & \ddots & \vdots \\ A_{n,1} & A_{n,2} & \dots & A_{n,d} \end{bmatrix} \quad (3)$$

$$M_{OA} = \begin{bmatrix} f([A_{1,1} & A_{1,2} & \dots & A_{1,d}]) \\ f([A_{2,1} & A_{2,2} & \dots & A_{2,d}]) \\ \vdots & \vdots & \ddots & \vdots \\ f([A_{n,1} & A_{n,2} & \dots & A_{n,d}]) \end{bmatrix} \quad (4)$$

In matrix (3), M_{Ant} is the matrix that contains the location of each ant, $A_{i,j}$ is the value of the *j*th variable (dimension) of the *i*th ant, *n* indicates the number of ants and, *d* shows the number of variables. The location of an ant is related to the parameters of a particular solution. To score each ant, a fitness function (target) is used during optimization, and the fitness values of every ant are stored in the matrix (4). Where M_{OA} indicates the matrix containing the fitness of each ant, and *f* is the target function. In addition to ants, it is assumed that the ant-lions are also hidden somewhere in the search space. To store the location and the fitness of ant-lions, the matrices (5), and (6) are used:

$$M_{Ant-lion} = \begin{bmatrix} ALi_{1,1} & ALi_{1,2} & \dots & ALi_{1,d} \\ ALi_{2,1} & ALi_{2,2} & \dots & ALi_{2,d} \\ \vdots & \vdots & \ddots & \vdots \\ ALi_{n,1} & ALi_{n,2} & \dots & ALi_{n,d} \end{bmatrix} \quad (5)$$

$$M_{OAL} = \begin{bmatrix} f([ALi_{1,1} & ALi_{1,2} & \dots & ALi_{1,d}]) \\ f([ALi_{2,1} & ALi_{2,2} & \dots & ALi_{2,d}]) \\ \vdots & \vdots & \ddots & \vdots \\ f([ALi_{n,1} & ALi_{n,2} & \dots & ALi_{n,d}]) \end{bmatrix} \quad (6)$$

Where $M_{Ant-lion}$ indicates the matrix containing the location of each ant-lion, $ALi_{i,j}$ is the value of the *j*th dimension of the *i*th ant-lion, *n* indicates the number of ant-lions, and *d* shows the number of variables (dimensions). In matrix (6), M_{OAL} indicates the matrix containing the fitness of each ant-lion, and *f* shows the target function.

3.2.2. Random movements of ants

The ants update their locations by randomly moving at each optimization step. Since all the search spaces have a border (variable range), a normalization procedure is applied to the random movements to keep them within the range of the search space. The method used here is Min-Max normalization:

$$X_i^t = \frac{(X_i^t - a_i) * (d_i^t - c_i^t)}{(b_i^t - a_i)} + c_i \quad (7)$$

Where b_i and a_i show the maximum and the minimum random movements of the *i*th variable, respectively. In addition, d_i^t and c_i^t indicate the maximum and the minimum of the *i*th variable in the *t*th iteration, respectively. This equation must be applied iteratively to ensure that the random movements are kept in the range.

3.2.3. Getting caught in the ant-lions traps

As discussed before, ant-lions traps affect the random movements of ants. To model this assumption mathematically, the following equations are used:

$$c_i^t = Ant - lion_j^t + c^t \quad d_i^t = Ant - lion_j^t + d^t \quad (8)$$

Where d^t and c^t show the vectors containing the maximum and the minimum values of all variables in the *t*th iteration, respectively. Moreover, d_i^t and c_i^t are the maximum and the minimum values of all variables of the *i*th ant, respectively. Also,

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1: Input: Search space, number of iterations (T), number of ants, fitness function, and number of ant-lions
2: Output: The elitist ant-lion and its fitness
3: Initialization
4: Initialize  $n$  ant-lion's positions and a population of  $n$  ant's positions randomly.
5: Calculate the fitness of all ant-lions, and ants.
6: Select the fittest ant-lion, and elite.
7: Set  $t = 0$ .
8: while The iteration number is less than the maximum iteration number do
9:   for all ( do Each ant )
10:    Select an ant-lion using roulette wheel
11:    Perform random walk of  $Ant_i$  around  $Ant - lion_A$  using Eq. (7);  $R_A^t$ 
12:    Perform random walk of  $Ant_i$  around  $Ant - lion_E$  using Eq. (7);  $R_E^t$ 
13:    Update the position of  $Ant_i$  using Eq. (11)
14:   end for
15:   Calculate the fitness of all ants.
16:   Merge the ant and ant-lions, sort them according to fitness and select the worst  $n$  as the ants (catching prey),
   and the best  $n$  agents as the next ant-lions.
17:   Update elite if an ant-lion takes fitter than the elite.
18: end while

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Fig. 2. The pseudo code of the ant-lion optimizer algorithm.

$Ant - lion_i^t$ indicates the location of the j th selected ant-lion in the t th iteration. These equations show that the ants move around the chosen ant-lion in a spherical cloud determined by the vectors c and d .

To model this behavior mathematically, the spherical cloud radius of the ant's random movements is adaptively reduced. The following equations are suggested:

$$c^t = \frac{c^t}{I} \quad d^t = \frac{d^t}{I} \quad (9)$$

Where I refers to a proportion, and d^t and c^t show the vectors containing the maximum and the minimum values of all variables in the t th iteration, respectively. In this equation, $I = 10^{wi} \times \frac{t}{T}$, where t shows the current iteration, T shows the maximum number of iterations, and wi indicates a constant that is determined based on the current iteration ($wi = 2$ when $t > 0.1T$, $wi = 3$ when $t > 0.5T$, $wi = 4$ when $t > 0.75T$, $wi = 5$ when $t > 0.9T$, and $wi = 6$ when $t > 0.95T$). The constant wi can adjust the value of accuracy. These equations diminish the radius of updating the ant's location and simulate the process of an ant slipping into the holes. This ensures the efficiency of the search space.

The final step of hunting is when an ant arrives at the bottom of the hole and it is grabbed by the ant-lion's jaw. After this, the ant-lion pulls the ant into the soil and eats it. To model this process, it is assumed that prey capturing is done whenever the ant has a higher fitness score than the corresponding ant-lion (drawn into the soil). The ant-lion must update its location based on its last trap, to increase the chance of capturing new prey. The Eq. (10) is written in this regard:

$$Ant - lion_j^t = Ant_i^t \quad \text{if} \quad f(Ant_i^t) > f(Ant - lion_j^t) \quad (10)$$

Where t represents the current iteration, $Ant - lion_i^t$ shows the location of the j th selected ant-lion in the t th iteration, and Ant_i^t is the location of the i th ant in the t th iteration.

3.2.4. Elitism

Elitism is an essential aspect of evolutionary algorithms, which allows them to keep the best-ever solutions obtained from the optimization process. In this study, the best-selected ant-lion in each iteration was saved and considered as an elite. For an ant-lion with a higher fitness value, it must be able to influence the activities of all ants during the iterations. Therefore, it is assumed that each ant is moving randomly around an ant-lion that is selected simultaneously based on the roulette wheel and the fitness value. This movement is described as follow:

$$Ant_i^t = \frac{R_A^t + R_E^t}{2} \quad (11)$$

Where R_A^t shows the random movement around the selected ant-lion based on the roulette wheel at the t th iteration, R_E^t shows the random movement around the elite ant-lion in the t th iteration, and Ant_i^t indicates the location of the i th ant in the t th iteration.

The Ant-lion optimizer in the following algorithm is formulated. The optimizer included the basic nature of ants in hunting the prey, and it involved five steps: (1) random walk, (2) building of traps, (3) trapping ants in the traps, (4) catching of the prey, and (5) reconstructing the traps (see Fig. 2).

4. Proposed method

A new approach to classifying network traffic using deep learning and a meta-heuristic algorithm is presented. A general view of the proposed method is shown in Fig. 3. The proposed method contains four phases described as follows. 1—Preprocessing of the traffic flows with entropy and variance of entropy. 2—Automated feature extraction using 1D-CNN. 3—Efficient feature selection using the ant-lion meta-heuristic algorithm. 4—New instances classification using a combined fuzzy-SOM based clustering. Each phase is explained in detail.

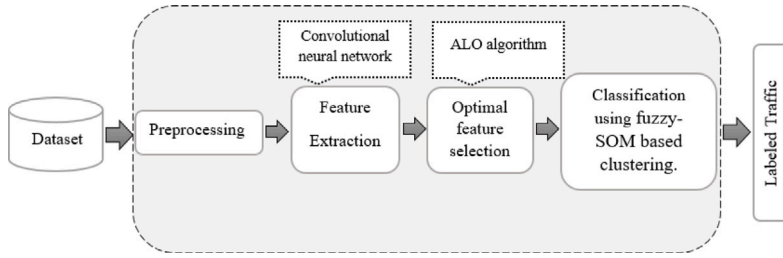


Fig. 3. A general view of the proposed traffic classification method.

4.1. Preprocessing

One critical and commonly done preprocessing step in deep neural networks is to normalize the data first. Therefore, the preprocessing is applied to the dataset, with the aim of normalization and standardization of the data. Moreover, it is important to identify the outliers as they can reduce the classification accuracy. Error-prone data of a flow is determined by calculating the entropy of the gathered statistical features values. The lower entropy proves an individual discipline in the data flow of traffic, while the higher value shows the chaos. Therefore, a flow that has a feature with low entropy probably contains a pattern redundancy over time. In other words, the pattern of this flow is more stable and can be considered as a good option for the training phase. The entropy value $E_{(i,w_t)}$ for a data sequence like $C_{(i,w_t)}$ at time slot w_t is calculated as Eq. (12):

$$E_{(i,w_t)} = -\log \frac{C_{(i,w_t)}}{\sum_{i=1}^n C_{(i,w_t)}} + \lambda_{(i,w_t)} \quad (12)$$

$$\lambda_{(i,w_t)} = \begin{cases} \left| \log \frac{c_{(i,w_{t+1})}}{c_{(i,w_t)}} \right|, & c_{(i,w_t)} \geq c_{(i,w_{t+1})} \\ \left| \log \frac{c_{(i,w_t)}}{c_{(i,w_{t+1})}} \right|, & c_{(i,w_t)} < c_{(i,w_{t+1})} \end{cases} \quad (13)$$

In addition to calculating the feature entropy in the traffic flow pattern, the variance of the entropy (Eq. (13)) of such values can be measured and used as an input variable.

4.2. Feature extraction

An important step in network traffic classification methods is feature extraction. Finding the desirable features that can lead to high accuracy is difficult and time consuming. Therefore, it is important to extract the features efficiently. The CNN is one of the most popular algorithms in deep learning for feature extraction. For this purpose, in the next step, the processed data are sent to a CNN to extract features. In this phase, a CNN algorithm was used because this method performs less preprocessing tasks compared to other feature extraction techniques. So, it needs data with similar sizes to map them into CNN. Therefore, a one-dimensional array of remaining statistical features from the previous stage was provided as input for the convolutional network.

The optimization of hyperparameters is an essential part of deep learning. In many cases, the selection of hyperparameters is done manually with trial and errors, which is a time-consuming method. Moreover, the correct selection of this type of parameters has a significant impact on the accuracy. Therefore, a technique that shifts the feature extraction operation from a manual to an automatic manner was proposed. The proposed method deals with an estimation model that uses the grid search algorithm to estimate the number of neurons, the number of iterations for training the network, and other parameters of the CNN. Table 1 provides the values of the hyperparameters obtained using the grid search algorithm.

This model was trained using the labels of the training data to recognize the class of each flow (the protocol generating the flow) based on the different combinations of the features created in the intermediate layers. The CNN model in this study contains two convolutional layers, two pooling layers, and five fully connected layers. The fully connected layers are the final classification layers, where the output features are calculated by multiplying the fully connected weight matrix by the input feature vector. The output of this phase is composed of several new combinational features that must be extracted from the hidden layers in the convolutional network structure.

4.3. Feature selection

Another problem with deep learning models is that they require large amounts of data to train. Since these models contain several hidden layers, the optimized adjustment of the weights cannot be made entirely and correctly if there is a small amount of data. According to other studies, the optimal and desirable outcome is not achievable without sufficient data. Therefore, lower accuracy can be achieved for applications with less data available.

Feature sets usually include dependent, redundant and correlated members that degrade the classification performance and increase the training time. To enhance the classification performance, irrelevant features must be removed in the feature selection

Table 1
The values of hyperparameters.

Parameter	Value
Activation	<i>softmax, Relu</i>
Epochs, batch size	100, 20
Optimizer	<i>Adam'</i>
Learning rate	0.2
Dropout	0.5
Loss	<i>categorical_crossentropy</i>

phase. In this paper, ALO was used as the search algorithm to find the optimal feature set and maximize the classification performance. In this regard, the use of the ant-lion meta-heuristic algorithms seemed beneficial to select optimal features and overcome the problem of lack of data. Meta-heuristic algorithms utilize some mechanisms and basic operations to finally achieve a proper and optimum solution through a set of iterations. After extracting the features, the ant-lion meta-heuristic algorithm was used to find an optimal subset of the extracted features with high accuracy. Here, each solution is defined as a vector with a size k . The values of this vector are binary, and indicate the presence or absence of a feature with the corresponding index in the optimized set. Eq. (14) is used to evaluate the suitability of the obtained solutions. The result of this equation showed the amount of the worthiness of a solution based on the size of the set, and the classification accuracy using the features selected by the SVM (Support Vector Machine). The target function for evaluating the ant-lion's fitness is described as follows:

$$F(S_i) = \frac{P(S_i)}{\#feature(s_i) + 1} \quad (14)$$

Where $P(S_i)$ indicates the accuracy obtained by the feature set sol_i using SVM, and $\#feature(S_i)$ shows the number of the members of the feature set in this solution. The output of this phase is a feature set with high accuracy in determining traffic classification.

4.4. Traffic detection

To classify the traffic based on the optimized features, a combination of self-organized map (SOM) clustering and fuzzy logic was used. Data might contain complex structures so that even extracting meaningful patterns using the best data mining techniques can be a challenging task. Clustering refers to dividing a dissimilar pattern into several similar subsets or groups. A SOM is an unsupervised neural network where neurons are organized in a standard network structure with a small number of parameters. Each neuron has an n -dimensional weight vector. Weight vectors (synapses) connect the weight vectors of the input layer to the output layer called the map or competition layer. The neurons are connected through a proximity function. Each input layer, based on the most similarity, activates a neuron in the output layer, called the winner cell. The most crucial difference between the SOM training algorithm and other vector measuring algorithms is that in addition to the single communication weight with the highest adaptation (the winner cell), the weight of the neighbor cells of the winner cell is also updated.

First, based on the training set, the values of the extracted features are divided into k clusters using clustering in the SOM network which is an unsupervised learning network. At the same time, the label of each flow is known according to the protocol values (UDP, HTTPS, etc.). The value of k is initially equal to $k = 6$; however as the number of clusters increases the partitioning of flow instances based on features similarities becomes more accurate. As a result, an improvement in the accuracy of the proposed method is expected. Each cluster might have several types of traffic and after performing clustering, the attributes of the feature center of a cluster are used as the indicators of the members of that cluster. By considering the traffic labels (types of protocols) in this training dataset, the number of each type of protocol in a cluster is determined and represented as a percentage of all members of that cluster (Eq. (15)). Therefore, a number in the range of $[0, 1]$ is obtained for each cluster, so that the summation of these numbers M_{ij} is equal to one (Eq. (16)).

$$m_{ik} = \frac{tv_{ik}}{n_i} \quad (15)$$

$$\sum_{i=1}^{\#cluster} m_{ij} = 1 \quad \forall j \in clusters \quad (16)$$

In Eq. (15), tv_{ik} is the number of the traffic types i in the cluster k , and n_i is the total number of traffic types in the cluster k . These values are stored as the model results to evaluate new traffic. Next, in the evaluation phase (test), the distance between each traffic instance l and the cluster centers is calculated as a_{lk} . In other words, it determines the distance between the features of each flow and the features in each cluster k . To this end, the Euclidean distance is used:

$$a_{lk} = \epsilon + \sum_{f=1}^{\#feature} \|center(k, f) - sample(l, f)\|^2 \quad (17)$$

Table 2

Data set specifications (the number of samples in different classes in dataset).

Class	CHAT	FT	MAIL	STREAMING	VOIP	P2P	VPN:CHAT	VPN:FT	VPN:MAIL	VPN:STREAMING	VPN:VOIP	VPN:P2P
Samples	67790	103801	31334	36128	169309	62817	69911	102560	53040	24719	113886	105063

Table 3

Accuracy of proposed method and 1D-CNN.

Dataset	Proposed method			CNN-1D		
	Accuracy	Recall	Precision	Accuracy	Recall	Precision
CHAT	0.97	0.97	0.96	0.94	0.94	0.96
FTP	1.00	0.99	0.99	0.98	0.98	0.97
MAIL	0.94	0.91	0.93	0.87	0.90	0.86
STREAMING	0.95	0.94	0.95	0.89	0.89	0.88
VOIP	1.00	1.00	1.00	1.00	0.99	0.99
P2P	0.99	0.97	0.96	0.96	0.98	0.95
CHAT:VPN	0.98	0.98	0.98	0.95	0.99	0.96
FT:VPN	1.00	1.00	0.99	0.99	0.98	1.00
MAIL:VPN	0.97	0.97	0.96	0.96	0.92	0.91
STREAMING:VPN	0.94	0.93	0.92	0.88	0.87	0.89
VOIP:VPN	1.00	1.00	1.00	1.00	0.97	0.98
P2P:P2P	1.00	0.99	1.00	0.99	0.97	0.99
Avg/total	0.98	0.97	0.97	0.95	0.95	0.94

So that, $\#feature$ is the number of the features of each traffic flow, $center(k, f)$ is the value of feature f in the cluster center k , and the variable $sample(l, f)$ is the value of feature f in traffic l . The formula of the membership value in the fuzzy logic type-2, and the probability of the flow type i in traffic l can be calculated as follows:

$$p_{li} = \sum_{k=1}^{\#cluster} \frac{m_{ik}}{a_{lk}} \quad (18)$$

Having the values of P_{li} (as the indicators to calculate the probability of the traffic type i in the flow l), we can rank the probability values and then select the higher probability. In this way, the nature of a flow can be evaluated using the flexibility of fuzzy logic as a probable number. To dynamically update the created model and adapt to changes happening in the traffic situations, the SOM clustering is performed on the observed data in every specified time period to obtain the centers of the K clusters.

5. Experimental results

To implement the proposed traffic classification method, the Tensorflow framework [21], and the Keras library [22] of the Python language were used. An Asus laptop with the Nvidia Geforce-930 MX GPU, windows 10 operating system, and 12 GB main memory served as hardware. In this section, the dataset, the model evaluation methods, and the analysis of the results and the problems around them are discussed.

5.1. Dataset

The traffic data collected by the Information Security Centre of Excellence (also called the “ISCX VPN-non VPN” traffic dataset) was used [14]. The dataset is publicly available for research purposes. This dataset includes different types of traffic, like browsing, VoIP, Streaming, Chat, Emails, etc. There are about 14 types of traffic in total in this dataset. The “ISCX VPN- non VPN” dataset is collected by the TCP DUMP software and has an approximate size of 28 GB. Details about available classes are listed in Table 2. This dataset contains 23 features for each flow instance. These 23 features are the statistical features on the packets of each flow, which include the number of the forwarded packets, the size of the packet sent, the shortest time between packets arrivals, the size of packets, etc.

5.2. Parameters and model evaluation methods

Traffic classification accuracy is the most important indicator for evaluating and comparing of the proposed method with other techniques proposed in this area. Three criteria were used in this study.

$$Recall = \frac{TP}{TP + FN} \quad (19)$$

$$Precision = \frac{TP}{FP + TP} \quad (20)$$

$$Accuracy = \frac{TN + TP}{FN + FP + TP + TN} \quad (21)$$

Table 4
Confusion matrices using the proposed method.

	CHAT	FTP	MAIL	STREAMING	VOIP	P2P	CHAT:VPN	FTP:VPN	Mail:VPN	STREAMING:VPN	VOIP:VPN	P2P:VPN
CHAT	67608	0	180	60	0	0	33	4	0	4	0	0
FTP	1	103768	17	0	1	6	158	98	0	98	1	0
MAIL	0	6	30832	5	0	20	0	2	15	2	0	5
STREAMING	60	20	0	35715	0	0	0	1	33	0	0	2
VOIP	0	0	9	201	169301	0	15	45	801	1	19	0
P2P	20	0	159	0	0	62784	0	0	0	45	0	0
CHAT:VPN	104	4	0	21	3	0	69693	0	22	0	0	7
FTP:VPN	0	0	0	0	0	4	0	102375	0	0	0	0
MAIL:VPN	0	0	81	4	0	0	9	9	532165	214	0	0
STREAMING:VPN	8	3	0	0	4	9	1	11	1	24334	0	0
VOIP:VPN	0	0	61	115	0	0	0	15	0	6	113862	0
P2P:VPN	7	0	5	7	0	0	2	0	3	15	4	105049

Where FP, TP, and FN show false positive, true positive, and false negative, respectively. In addition, there are several methods for evaluating models. Generalizability is one of the necessary features for a model. This means that the created model can also show excellent performance on the new and unobserved instances. Another reason to use evaluation methods is the common problems in most datasets namely bias, and overfitting. The cross-validation method was used to solve these problems. In this technique, the dataset is partitioned into k different subsets and with each time one of these subsets is selected for testing, and other $(k - 1)$ remaining subsets are used for training. This process is repeated K times, and all subsets are used exactly once for testing and once for training. At the end, the average result of these K repetitions is considered as a final estimation. In this study, K was equal to 10 and the K -fold cross-validation was conducted randomly; therefore, the dependency between data and error level was decreased.

5.3. Evaluation results

This section presents the results obtained by applying the proposed method to recognize, and classify the network traffic. To perform a better comparison, along with evaluating the proposed method, a deep learning network with a dataset was also evaluated. This deep learning network is a 1D-CNN with two convolutional layers, two pooling layers, and multiple fully connected layers that performs feature extraction and traffic classification sequentially. In fact, the problem is to classify 12 classes of network applications. These classes contain six normal classes, and six VPN classes.

The results of applying the proposed method and the CNN to the dataset for identification and classification of traffics are provided in Table 3. The experiments were repeated several times, and the average accuracy of classifying the available classes and instances in each class was calculated according to accuracy, precision, and recall criteria. It reached an average accuracy of 98% on the dataset for a 12-classes problem. The CNN also achieved an average accuracy of 95% on the same dataset. Both methods have excellent performance and accuracy. As shown in Table 3, amongst all classes, the accuracy for VPN-Streaming, Streaming, and Mail classes is lower with the CNN method. This low accuracy for such applications is due to the availability of fewer training datasets compared to other classes. Because of this, CNN could not learn sufficient features to recognize the data and, consequently, the traffic. To yield successful results, artificial neural networks (ANNs) such as CNN depends on factors like the system architecture, the training algorithm, and the features utilized in the training. These factors ensure the design an optimal ANN. The proposed method used deep learning models to extract features automatically for network traffic classification and ant-lion meta-heuristic algorithm to solve the problem of insufficient data availability and selecting optimal features. Such a method must be able to classify different types of traffic with high accuracy and acceptable performance.

The proposed method achieved good accuracy even for the classes with fewer instances to train. This is due to the application of an evolutionary algorithm and its combination with a deep learning network, resulting in an optimized solution. Another privilege of the proposed method in traffic classification was the proper detection of applications like P2P and VoIP with high accuracy. The P2P data are usually encrypted with dedicated protocols, and recognizing them is too hard. Nevertheless, the proposed method can recognize such applications with high accuracy. In the best case, the method reached both the recall and the precision of 97% for all data, including the VPN traffics. The numerical form of the confusion matrix of the proposed method is shown in Table 4. The rows of this matrix correspond to the real classes of the instances and the columns show the predicted labels for instances.

5.4. Comparison with other methods

The proposed method was compared with four deep learning algorithms. These deep learning algorithms are: convolutional neural networks (CNN) [18], Long short-term memory (LSTM) [23], Deep belief network (DBN) [13], and Recurrent neural network (RNN) [24]. Experimental results showed that DBN-based traffic classification had higher classification accuracy for

Table 5

The comparison between our proposed method and other deep learning methods which have implemented by us on the “ISCX VPN-non VPN” dataset.

Method	Precision	Recall	Accuracy
Deep belief network (DBN) [13]	95%	96%	96%
Long short-term memory (LSTM) [23]	92%	93%	92%
Recurrent neural network (RNN) [24]	94%	93%	94%
Convolutional neural network (CNN) [18]	94%	95%	95%
Proposed method	97%	97%	98%

Table 6

The accuracy comparison between the proposed method and other existing methods [14], [15], [17], and [18] on the “ISCX VPN-non VPN” dataset.

Paper	Method	Accuracy
Gil, et al. [14]	Decision tree	92%
Yamansavascular et al. [15]	Knn	94%
Lotfollahi et al. [17]	CNN, SAE	94%
Wang et al. [18]	CNN	92%
Proposed method	CNN-ALO	98%

P2P applications. DBN, LSTM, RNN, and CNN were used for comparison purposes. Table 5 indicates that the proposed method outperformed four alternative algorithms in the traffic characterization task for the “ISCX VPN-nonVPN” dataset. Similarly, these comparisons verify the power of the proposed method for the network traffic classification. The average accuracy obtained by the proposed method reached 98%.

5.5. Discussion

In order to evaluate the proposed method as accurately as possible, a comparison with other related work was carried out. The followings are some studies that have used the same dataset as this study (e.g., ISCX VPN-non VPN). Several papers have been published to classify network traffic using machine learning methods. Two of the essential published papers that used the “ISCX VPN-non VPN” dataset, focused on machine learning methods. Gil et al. [14] used temporal features like flow arrival time, the number of flow Bytes per second, etc., for network traffic classification. They also applied a decision tree and the nearest neighbor algorithm to classify the traffic. The authors achieved an accuracy of 92% in recognizing six classes, including chat, email, FTP, ..., and normal classes by utilizing the decision tree algorithm. By comparing the results of this paper with the results obtained here, the superiority of our method in classification becomes quite clear. Yamansavascular et al. [15], achieved 94% accuracy with the same dataset, 111 flow features with 12 application classes, and KNN algorithm. The main drawback of these methods is that an expert is required to perform the feature extraction and selection processes. Therefore, such methods are prone to human error, time-consuming, and expensive.

Recently, some studies have applied deep learning ideas to the “ISCX VPN-non VPN” dataset. In [16], Rezaei et al. introduced a general framework to classify the network traffic based on deep learning. They also discussed the problems and challenges of their proposed method. Lotfollahi et al. [17], introduced a model based on CNN and SAE for classifying encrypted traffic. The number of classes and data instances in their work were almost the same as in the present study. They achieved an accuracy of 94% in traffic classification by the CNN model, while our proposed method reached an accuracy above 95% for most applications and the accuracy of 98% on average. Their proposed network could not achieve a good improvement in detecting different traffics like streaming and mail. The average accuracy for these classes is about 87%, which is due to the lack of a sufficient training dataset for these types of traffic. However, the proposed method reached an excellent accuracy of 95% by combining deep learning networks and optimized evolutionary algorithms for traffics with insufficient training sets. Wang et al. [18], proposed classifying encrypted traffic method using the one-dimensional convolutional neural network. This method combines feature extraction, feature selection, and the classifier in a unified final framework for automated learning of the non-linear relation between row input and expected output. The proposed method was applied to the “ISCX VPN-non VPN” dataset and obtained an accuracy of 92% for 12 classes.

As discussed earlier, one of the challenges in deep learning networks is the use of hyperparameters in a random and trial and error manner. This challenge is due to the complexity of neural network configuration and the many parameters that need to be adjusted. The correct and optimized selection of these parameters has a significant impact on increasing accuracy and reducing execution time, especially for convolutional networks that are very sensitive to this factor. Therefore, to solve this problem, an estimation or predictive model is used that works on the grid search method and estimates several parameters like the number of neurons, the number of iterations to train the network, the activation function, the cost function, etc. A general comparison between the proposed method and other techniques focused on the “ISCX VPN-non VPN” dataset is presented in Table 6. The proposed method reached a higher accuracy compared to both machine learning and deep learning methods. Since the feature extraction is automatically performed in the deep learning network, many features might be extracted which might be ineffective in the accuracy and even increase the computational cost and time.

6. Conclusions

A method based on the evolutionary algorithm and deep learning was proposed to automatically select features and accurately classify network traffic. Several studies have recently focused on deep learning methods to extract features automatically. However, deep learning models need massive training data that is not always available. Since these models are composed of multiple hidden layers, the optimized weight adjustment cannot be done accurately and correctly with insufficient training data. Moreover, the initial point for adjusting weight is essential. If the initial network settings are insufficient, there is a risk that the optimization of the weights will stop at a local optimum. To address this problem, an evolutionary algorithm was used. This algorithm creates one or more random solutions and searches for a global optimum in the search space. In addition, the cost of recruiting a human expert for detecting and recognizing the traffic features was eliminated by using the automated feature extraction method. The proposed method benefits from an evolutionary algorithm to find an optimal solution and select the appropriate features, reducing the execution time and also the computational cost. This method is a combination of CNN, ALO, and SOM, and its purpose is to classify network traffic as accurately as possible with any size of the training dataset. By combining an evolutionary algorithm and neural networks, the proposed method can find an optimized training approach to classify network traffic with high accuracy. The results showed that the suggested method outperformed all similar techniques on the “ISCX VPN-non-VPN” dataset.

CRedit authorship contribution statement

Saadat Izadi: Conception and design of study, Acquisition of data, Analysis and/or interpretation of data, Writing – original draft, Writing – review & editing. **Mahmood Ahmadi:** Conception and design of study, Acquisition of data, Analysis and/or interpretation of data, Writing – original draft, Writing – review & editing. **Rojia Nikbazzm:** Conception and design of study, Acquisition of data, Analysis and/or interpretation of data, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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